

$$\frac{\partial E_{in}(w, b)}{\partial w_j} = \frac{1}{n} \sum_{i=1}^n \left(\frac{a^{(i)}}{\tilde{a}^{(i)}} - \frac{1-a^{(i)}}{1-\tilde{a}^{(i)}} \right) \tilde{a}^{(i)} (1-\tilde{a}^{(i)}) x_j^{(i)}$$

$$\frac{\partial E_{in}(w, b)}{\partial w_j} = \frac{\partial}{\partial w_j} \frac{1}{n} \sum_{i=1}^n [-a^{(i)} \ln(\tilde{a}^{(i)}) - (1-a^{(i)}) \ln(1-\tilde{a}^{(i)})]$$

$$= \frac{1}{n} \sum_{i=1}^n - \frac{a^{(i)}}{\tilde{a}^{(i)}} \frac{\partial \tilde{a}^{(i)}}{\partial w_j} + \frac{1-a^{(i)}}{1-\tilde{a}^{(i)}} \frac{\partial \tilde{a}^{(i)}}{\partial w_j}$$

$$= \frac{1}{n} \sum_{i=1}^n [a^{(i)} (1-\tilde{a}^{(i)}) + (1-a^{(i)}) \tilde{a}^{(i)}] x_j^{(i)}$$

$$= \frac{1}{n} \sum_{i=1}^n [a^{(i)} + \tilde{a}^{(i)} - a^{(i)} \tilde{a}^{(i)} - a^{(i)} \tilde{a}^{(i)}] x_j^{(i)}$$

$$\frac{\partial E_{in}(w, b)}{\partial w_j} = \frac{1}{n} \sum_{i=1}^n (a^{(i)} - \tilde{a}^{(i)}) x_j^{(i)}$$

$$w \leftarrow w - \frac{1}{n} x^T (A - \tilde{A})$$

$$b \leftarrow b - \frac{1}{n} \sum_{i=1}^n (a^{(i)} - \tilde{a}^{(i)})$$

$$w^*, b^* = \arg \min_{w, b} E_{in}(w, b)$$

regularized

$$\text{Bayes } \sum_{j=1}^n w_j^2 \leq C$$

$$w_{reg}^*, b_{reg}^* = \arg \min [E_{in}(w, b) + \frac{\lambda}{n} \sum_{j=1}^n w_j^2]$$

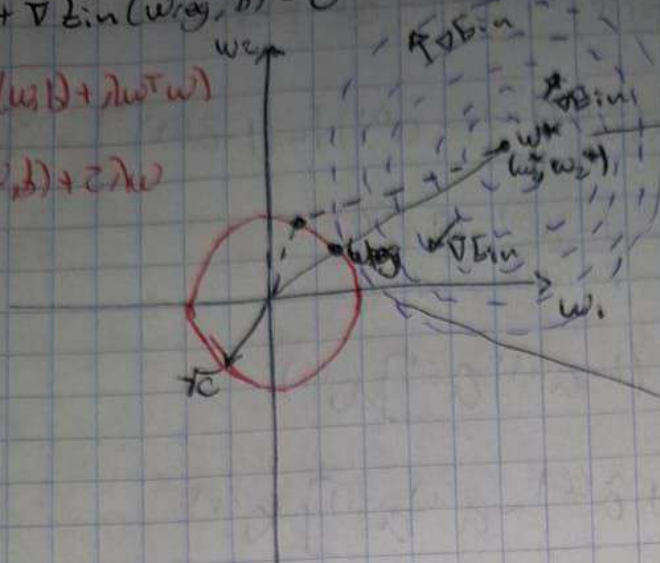
$$\frac{\partial}{\partial w_j} [E_{in}(w, b) + \frac{\lambda}{n} \sum_{j=1}^n w_j^2] = \frac{1}{n} \sum_{i=1}^n (a^{(i)} - \tilde{a}^{(i)}) x_j^{(i)} + \frac{2\lambda}{n} w_j$$

$$w_{reg}^*, b_{reg}^* = \arg \min [E_{in}(w, b) + \frac{\lambda}{n} \text{reg}(w)]$$

$$2\lambda w_{reg} + \nabla E_{in}(w_{reg}, b) = 0$$

$$\nabla_w (E_{in}(w, b) + \lambda w^T w)$$

$$\nabla_w E_{in}(w, b) + 2\lambda w$$



$$w^*, b^* = \arg \min_{w, b} E_{in}(w, b)$$

bajo

$$\sum_{j=1}^n w_j^2 \leq C$$

$$w^T w \leq C$$

$$\|w\|_2^2 \leq C$$

$$w_1^2 + w_2^2 \leq \sqrt{C}^2$$

$$w_{reg} = -K \nabla_w E_{in}(w, b)$$

Árboles de decisión

~~hacer~~ $f: X \rightarrow Y$

y quisiéramos ajustar

$h: X \rightarrow Y$ a partir de $D = \{(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)})\}$

$$\begin{matrix} & f_1 & f_2 & \dots & f_n \\ \begin{matrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(n)} \end{matrix} & \begin{bmatrix} x_1^{(1)} \\ x_1^{(2)} \\ \vdots \\ x_1^{(n)} \end{bmatrix} & \begin{bmatrix} x_2^{(1)} \\ x_2^{(2)} \\ \vdots \\ x_2^{(n)} \end{bmatrix} & \dots & \begin{bmatrix} x_n^{(1)} \\ x_n^{(2)} \\ \vdots \\ x_n^{(n)} \end{bmatrix} \end{matrix} \quad \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{bmatrix}$$